



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Smart Medical Inventory and Equipment Management System

CSA0206 - C-programming

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INTRODUCTION

- Tracks medicines, stock levels, expiry dates, and medical equipment in a centralized system.
- Provides real-time inventory updates to maintain accurate stock records.
- Generates alerts for low stock and medicines nearing expiry to prevent shortages and wastage.
- Monitors the availability and maintenance of medical equipment for efficient healthcare operations.
- Reduces manual errors, improves inventory management, and enhances patient care through automation.



LITERATURE REVIEW

Authors& Year	Title	Method	Finding	Gap
Li et al. (2024)	Smart Hospital Inventory System	Multi-Agent Reinforcement Learning	Improved medicine availability and reduced inventory costs.	No medical equipment tracking.
Walczyk et al. (2024)	Automated Drug Inventory	Inventory model comparison	Reduced expired medicines and improved stock control.	Focused only on medicine cabinets.
Bekele et al. (2025)	Essential Medicine Inventory	Cross-sectional study	Improved stock management and reduced stock-outs.	Did not include equipment management.
Abu Zwaida et al. (2025)	AI-Based Drug Inventory	Deep Reinforcement Learning	Optimized medicine stock and minimized shortages.	Expiry monitoring and equipment tracking were not included.
MDPI Review (2026)	AI in Healthcare Inventory	Systematic Literature Review	AI improved demand forecasting and inventory efficiency.	Lacks a fully integrated system for medicines and equipment.



PROBLEM STATEMENT

- Manual inventory management leads to inaccurate records and frequent human errors.
- Expired medicines may remain in stock due to the lack of proper expiry tracking.
- Medicine shortages or overstocking occur because stock levels are not monitored effectively.
- Medical equipment is difficult to track, resulting in misplacement, unavailability, or delayed maintenance.
- Lack of real-time monitoring reduces operational efficiency and can affect the quality of patient care.



OBJECTIVES

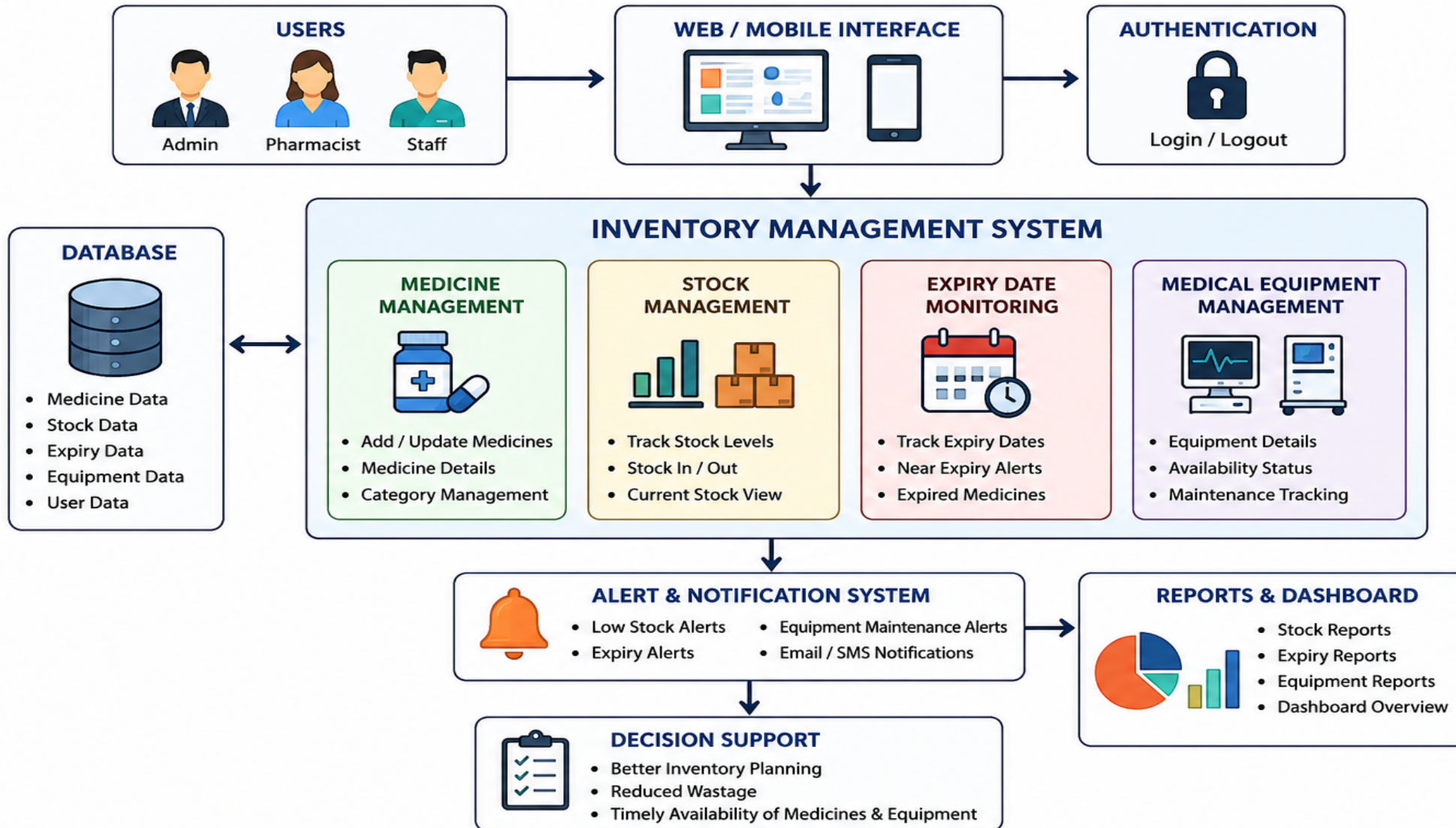
- To develop a centralized system for managing medicines and medical equipment.
- To monitor medicine stock levels in real time and maintain accurate inventory records.
- To track medicine expiry dates and generate alerts to reduce wastage.
- To monitor the availability and maintenance status of medical equipment.
- To improve inventory management by reducing manual errors and ensuring timely availability of medicines and equipment.



Engineer to Excel

SYSTEM ARCHITECTURE

MEDICINE AND MEDICAL EQUIPMENT TRACKING SYSTEM





Module-1: Requirement Analysis Module

- Collect user requirements from hospitals, pharmacies, and administrators.
- Analyze the existing inventory system to identify current problems and limitations.
- Define system requirements for medicine, stock, expiry, and equipment tracking.
- Identify hardware and software requirements needed for system development.
- Prepare the Software Requirements Specification (SRS) to guide the design and implementation of the system.



Module-2:Processing and display Module

- Process medicine and medical equipment data entered by the user.
- Update stock levels automatically when items are added or issued.
- Monitor expiry dates and identify near-expiry or expired medicines.
- Display real-time inventory status through a user-friendly dashboard.
- Generate alerts and notifications for low stock, expiry, and equipment maintenance.
- Produce reports for medicines, stock availability, expiry status, and medical equipment.



Module-3: Inventory Monitoring and Report Generation



- Monitor medicine stock levels continuously to ensure adequate availability.
- Identify low-stock, out-of-stock, and expired medicines for timely action.
- Track the availability, usage, and maintenance schedule of medical equipment.
- Generate inventory reports, stock summaries, and equipment status reports.
- Provide alerts for medicine restocking, expiry dates, and equipment servicing.
- Maintain accurate inventory records to support efficient hospital and pharmacy management.



CONCLUSION

- Provides an efficient and centralized system for managing medicines and medical equipment.
- Ensures accurate tracking of stock levels, expiry dates, and equipment availability.
- Reduces manual errors, medicine wastage, and inventory shortages through automation.
- Improves operational efficiency by generating real-time alerts, reports, and inventory updates.
- Enhances healthcare services by ensuring the timely availability of medicines and medical equipment for better patient care.

Thank
you!